

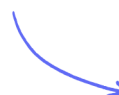
Structured Questions

Physical Chemistry Core Practicals

Molar Volume of a Gas / Determining Enthalpy Change of Reaction / Determining Concentrations / Standard Solutions

Medium (6 questions)	/66
Hard (1 question)	/10
Total Marks	/76

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Medium Questions

1 (a) This question is about ammonium chloride, NH_4Cl , a soluble ionic compound.

An aqueous solution of NH_4Cl contains both ammonium ions, NH_4^+ , and chloride ions, Cl^- .

i) State what would be **seen** on the addition of acidified silver nitrate solution to an aqueous solution of NH_4Cl .

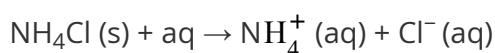
(1)

ii) Describe a test to confirm the presence of NH_4^+ ions in a solution of NH_4Cl . Include the result of the positive test.

(2)

(3 marks)

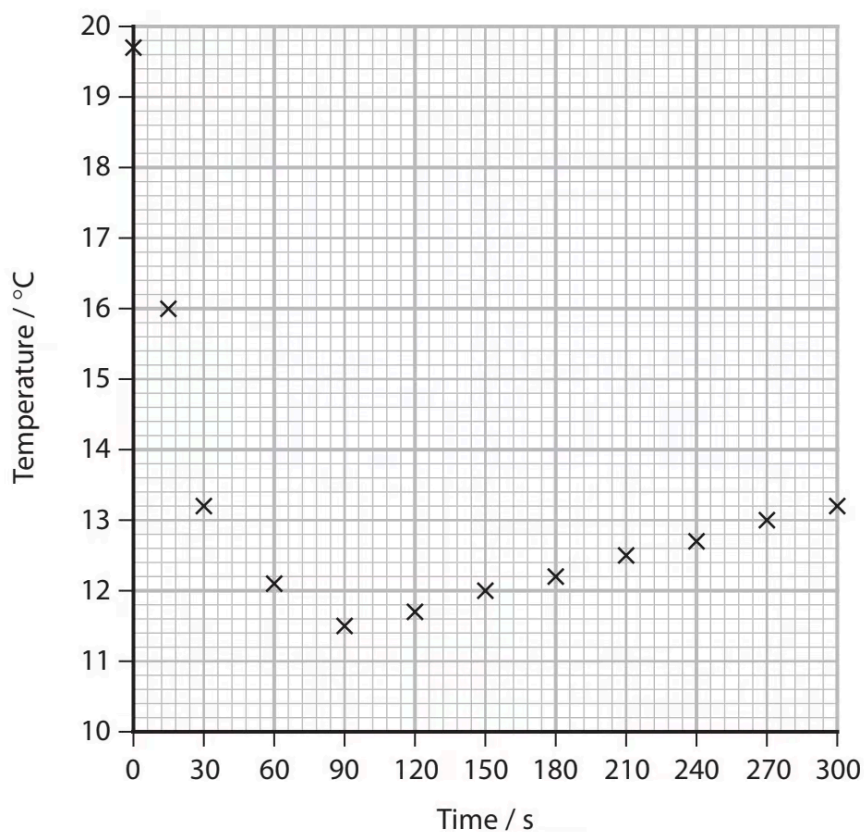
(b) A student investigated the enthalpy change when dissolving NH_4Cl in excess water.



Procedure

Step 1	Accurately weigh 7.17 g of NH_4Cl into a glass beaker.
Step 2	Fill a 50 cm^3 measuring cylinder with deionised water. Measure the temperature of the water using a thermometer.
Step 3	Pour the water from the measuring cylinder into the beaker and at the same time start a stopwatch. Stir the solution in the beaker, using the thermometer.
Step 4	Record the temperature at 15 s, 30 s and then at 30 s intervals while continuing to stir the solution.

The data from the experiment are shown on the graph.



i) Give **two** reasons why the student stirred the solution in Steps **3** and **4**.

(2)

ii) Use the graph to determine the maximum temperature change, ΔT , in this experiment. You **must** show your working on the graph.

(2)

iii) Another student carried out the experiment using a polystyrene cup in place of the glass beaker.

Explain how this student's graph would be different. You may annotate the graph as part of your answer.

(3)

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(7 marks)

- (c) The experimental results of another student were used in the equation shown to calculate the enthalpy change, ΔH , for dissolving one mole of NH_4Cl in excess water.

$$\Delta H = \frac{m \times c \times \Delta T}{n} = +14500 \text{ J mol}^{-1}$$

In the equation

m = mass of solution = 50 g

c = specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$

ΔT = maximum temperature change of solution

n = moles of NH_4Cl

i) State **two** assumptions made in this calculation. You do **not** need to justify your answers.

(2)

ii) The total percentage uncertainty in this experiment was 2.6%. Show that the enthalpy change of 14.5 kJ mol^{-1} is consistent with a data book value of 14.8 kJ mol^{-1}

(2)

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(4 marks)

- 2 (a)** Sodium hydroxide solution reacts with carbon dioxide in the air and should be standardised before use. Ethanedioic acid may be used for this standardisation.

A standard solution of ethanedioic acid, $(\text{COOH})_2$, is prepared.

- 2.40 g of solid ethanedioic acid is dissolved in approximately 100 cm^3 of deionised water in a beaker.
- The solution is transferred into a 250.0 cm^3 volumetric flask and made up to the mark with deionised water.

i) Give a possible reason why any solution remaining in the beaker is washed into the volumetric flask before making up to the mark.

(1)

ii) Calculate the concentration of this standard solution of ethanedioic acid in mol dm^{-3} .

Give your answer to an appropriate number of significant figures.

[Molar mass of ethanedioic acid = 90.0 g mol^{-1}] **(2)**

(3 marks)

- (b)** A **different** standard solution of ethanedioic acid is used to determine the concentration of a sodium hydroxide solution J.

Procedure

Step 1	A burette is rinsed with deionised water.
Step 2	The burette is then rinsed with $0.0900 \text{ mol dm}^{-3}$ ethanedioic acid and filled with this acid solution.
Step 3	A pipette is used to transfer 25.0 cm^3 portions of solution J to conical flasks.
Step 4	The portions are titrated with the ethanedioic acid solution using phenolphthalein indicator.

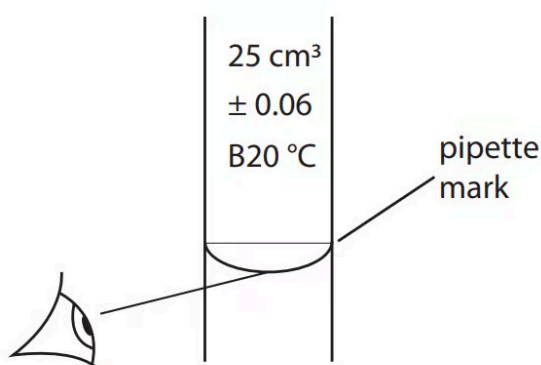
i) Explain why the burette is rinsed with ethanedioic acid solution in Step **2**.

(1)

ii) The diagram shows how the student read the filled pipette in Step **3**.

Identify the **two** mistakes the student made.

(2)



iii) The student completely emptied the pipette for each transfer in Step **3**.

Explain the effect **on the titre** of completely emptying the pipette rather than leaving a small amount of solution in the tip.

(2)

iv) State the colour change in the conical flask at the end-point.

(2)

Fromto

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(7 marks)

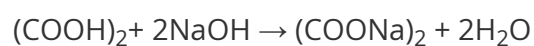
(c) The titration results are shown.

Titration	1	2	3
Final reading / cm ³	25.05	26.60	25.50
Initial reading / cm ³	0.00	2.00	1.00
Titre / cm ³			
Titres used in calculation of mean			

i) Complete the table and calculate the mean titre.

(2)

ii) Calculate the concentration of the sodium hydroxide solution in mol dm^{-3} .
The equation for the titration is



(3)

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(5 marks)

- 3 (a)** A student carried out experiments to determine the enthalpy change for the hydration of anhydrous copper(II) sulfate, CuSO_4 , to form hydrated copper(II) sulfate crystals, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

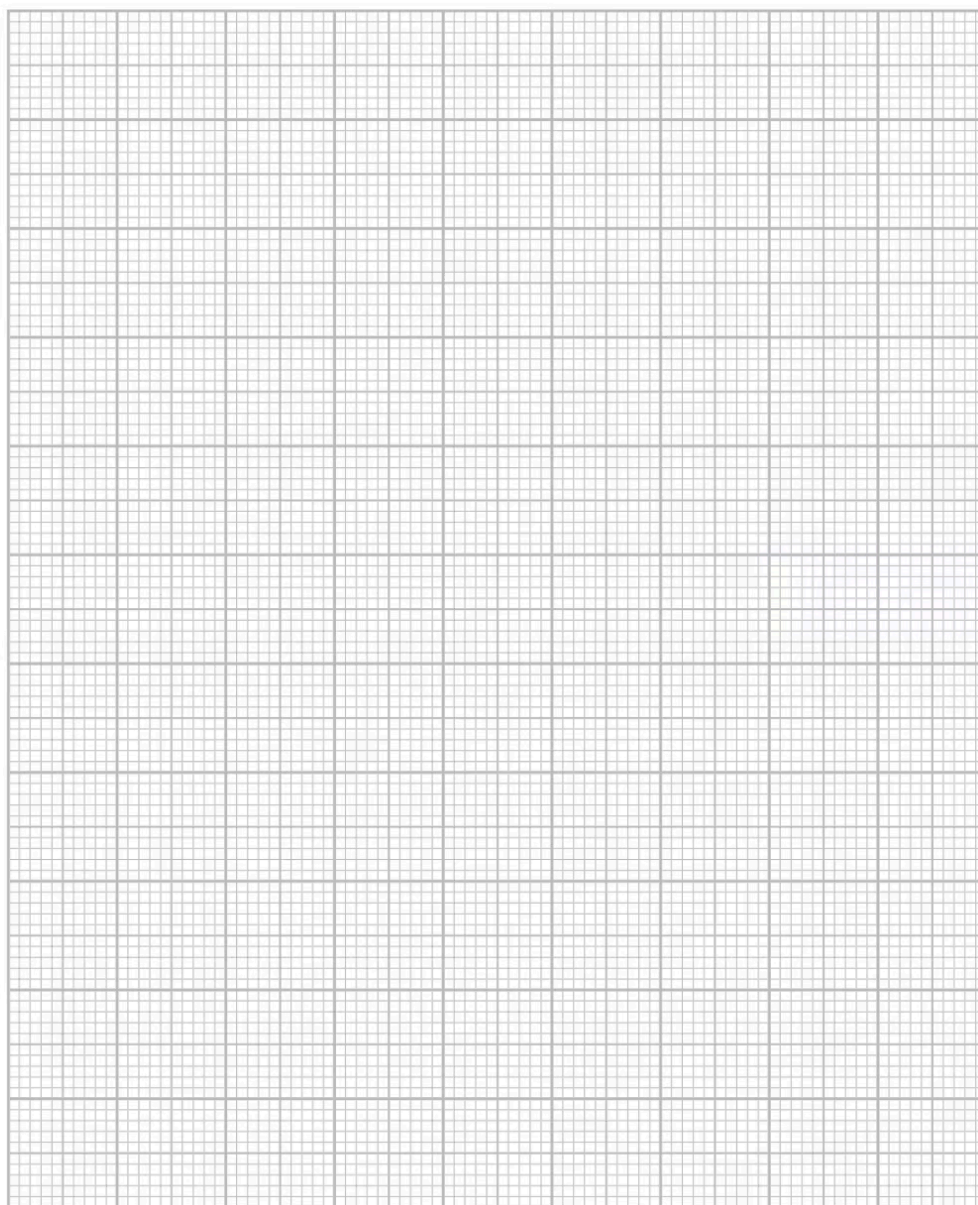
To find the enthalpy change of solution of anhydrous copper(II) sulfate, 25.0 cm^3 of distilled water was placed in a polystyrene cup and the temperature measured at one minute intervals.

After 2.5 minutes, 7.50 g of anhydrous copper(II) sulfate was added and the mixture stirred continuously.

The results are shown.

Time/minutes	0	1	2	2.5	3	4	5	6	7	8
Temperature/ $^{\circ}\text{C}$	21.1	21.0	21.0	X	34.2	37.6	36.9	36.1	35.2	34.3

Plot a graph of temperature against time on the grid.



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(3 marks)

(b) Determine the maximum temperature change, ΔT , using your graph.

You **must** show your working on the graph.

$\Delta T = \dots\dots\dots$

.....
.....
(2 marks)

(c) The value of the enthalpy change from this experiment was $-39.0 \text{ kJ mol}^{-1}$.

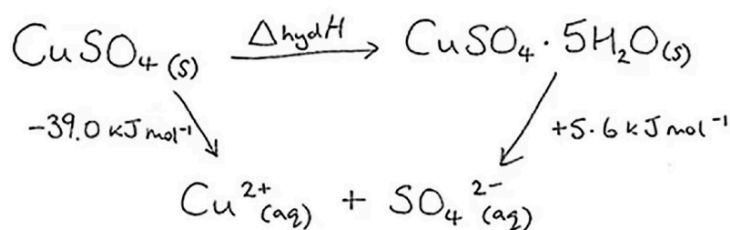
Give **one** possible reason why this value is different from a data book value of $-61.4 \text{ kJ mol}^{-1}$

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(1 mark)

(d) After another experiment to find the enthalpy change of solution of hydrated copper(II) sulfate crystals, the student constructed the Hess cycle shown.

i) Calculate the enthalpy change of hydration for the conversion of anhydrous copper(II) sulfate to hydrated copper(II) sulfate crystals.

(1)



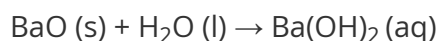
ii) Give **one** possible reason why the enthalpy change of hydration in (d)(i) could **not** be found directly by experiment.

(1)

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(2 marks)

- 4 (a) A saturated solution of barium hydroxide was formed by adding barium oxide to water until no more would dissolve. The equation for the reaction is



The resulting mixture was filtered to remove excess solid.

The concentration of the barium hydroxide solution was found by titrating portions of the saturated solution with hydrochloric acid of known concentration.

10.0 cm³ portions of the saturated barium hydroxide solution were placed in conical flasks and titrated with 0.200 mol dm⁻³ hydrochloric acid added from a burette.

Three drops of methyl orange indicator were added to the solution in each conical flask.

State the colour **change** observed at the end-point of the titration.

From to

(2 marks)

- (b) Some of the results are shown.

Titration	1	2	3	4
Final burette reading/ cm ³	22.60	44.45	23.05	
Initial burette reading/ cm ³	0.10	22.60	1.25	23.20
Titre/ cm ³	22.50	21.85		21.90

- i) Complete the table.

(1)

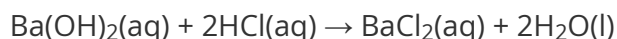
ii) Give a reason why the first titre should **not** be used to calculate the mean titre.

(1)

iii) Calculate the number of moles of hydrochloric acid in the mean titre.

(2)

iv) The equation for the reaction in the titration is



Calculate the concentration of barium hydroxide, in g dm^{-3} , giving your answer to an appropriate number of significant figures.

(3)

(7 marks)

- (c) Solid samples of soluble barium compounds such as barium oxide are toxic by inhalation due to the presence of barium ions.

Give a safety precaution that should be used to minimise this risk when adding barium oxide to water.

(1 mark)

(d) Barium also forms a peroxide. A bottle of barium peroxide has the hazard symbol .

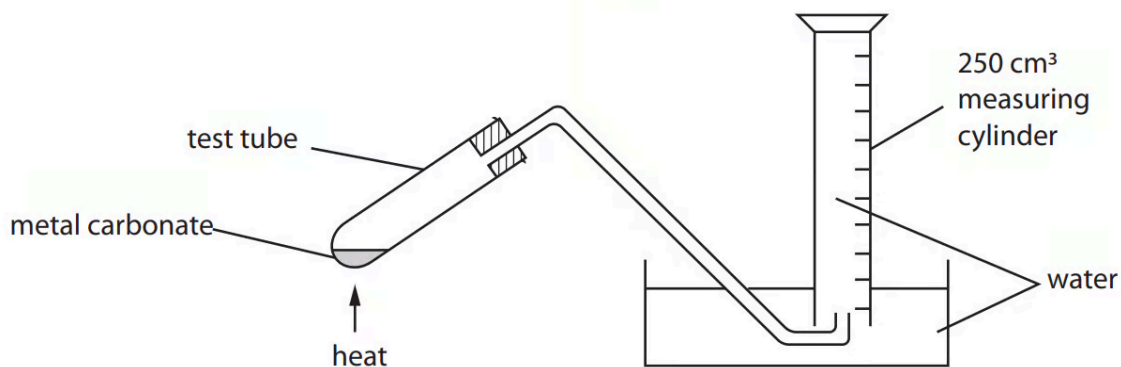


Give the meaning of this symbol.

(1 mark)

5 (a) This question is about the thermal decomposition of Group 2 carbonates.

A student heated a sample of a Group 2 carbonate until no more gas was produced. The equation for the decomposition is



Give a reason why the delivery tube must be removed from the water bath before removing the test tube from the heat source.

(1 mark)

(b) The results of the experiment are shown.

Measurement	Value
Volume of carbon dioxide / cm ³	95
Mass of test tube + carbonate / g	21.69
Mass of test tube / g	21.36
Mass of carbonate / g	0.33

i) Using the results of the experiment identify the Group 2 metal.

[Molar volume of gas at room temperature and pressure = $24.0 \text{ dm}^3 \text{ mol}^{-1}$]

(3)

ii) The student suggested that the experiment could be made more accurate by increasing the mass of carbonate from 0.33 g to 1.00 g.

No changes to the size of the apparatus or the method of measurement of the gas produced would be made.

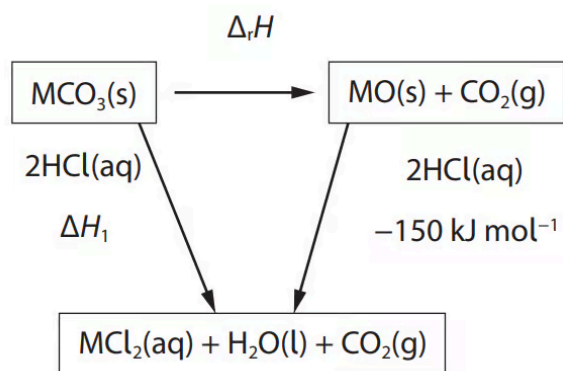
Comment on this suggestion.

(2)

(5 marks)

(c) The enthalpy change for the thermal decomposition of a carbonate, $\Delta_r H$, is difficult to measure directly.

An example of a Hess's Law cycle to determine it indirectly is



In an experiment to determine ΔH_1 , 0.050 mol of MCO₃ was placed in a 100 cm³ beaker. 60 cm³ of 2 mol dm⁻³ hydrochloric acid (an excess) was added and the mixture stirred. The maximum temperature rise measured was 6.0 °C. [Heat capacity of solution produced = 4.18 J g⁻¹ °C⁻¹ Density of solution = 1.0 g cm⁻³]

i) Calculate the enthalpy change, ΔH_1 , for the reaction between MCO₃ and hydrochloric acid in kJ mol⁻¹. Include a sign with your answer.

(2)

ii) Using your answer to (c)(i), calculate the enthalpy change, $\Delta_r H$, for the thermal decomposition of this Group 2 carbonate in kJ mol⁻¹.

Include a sign with your answer.

(1)

(3 marks)

- 6 (a)** A technician found a bottle of sodium hydroxide solution at the back of a cupboard. The technician determined its concentration by titrating 25.0 cm^3 samples against $0.500 \text{ mol dm}^{-3}$ hydrochloric acid from a burette.
- The results obtained are shown.

Complete the titre values.

		Titration number			
	Rough	1	2	3	4
Final reading / cm^3	24.90	21.25	42.85	21.80	43.15
Initial reading / cm^3	2.30	0.00	21.25	0.50	21.80
Titre/ cm^3					21.35

(1 mark)

- (b)** i) State why the value from Titration 2 was **not** used to calculate the mean.

(1)

- ii) Calculate the concentration of the sodium hydroxide solution in mol dm^{-3} .

(4)

(5 marks)

- (c) Each reading of the burette has an uncertainty of $\pm 0.05 \text{ cm}^3$

Calculate the percentage uncertainty in Titration 4.

(1 mark)

- (d) State the colour change that would be seen at the end-point in this titration using phenolphthalein as the indicator.

From to

(2 marks)

Hard Questions

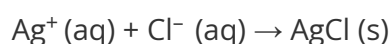
- 1 (a)** A student used a precipitation titration to determine the value of **x** in the formula of a sample of hydrated barium chloride, $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

Procedure

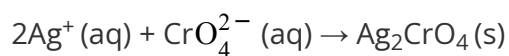
Step 1	Prepare a solution by dissolving 1.57 g of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ in deionised water, making the solution up to the mark in a 250.0 cm^3 volumetric flask and then mixing thoroughly.
Step 2	Use a pipette to transfer 10.0 cm^3 of the barium chloride solution into a conical flask. Add excess sodium sulfate solution and swirl the mixture.
Step 3	Fill a burette with $0.0324 \text{ mol dm}^{-3}$ silver nitrate solution.
Step 4	Add three drops of potassium chromate(VI) solution to the conical flask and titrate the contents, while swirling, with the silver nitrate solution. The end-point is shown by the appearance of a permanent pale red precipitate.
Step 5	Repeat Steps 2 to 4 until concordant results are obtained.

During the titration, two precipitation reactions occur.

Reaction 1	Silver ions react with chloride ions forming silver chloride.
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Reaction 2	Once all chloride ions have reacted, silver ions react with chromate(VI) ions to form a red precipitate of silver chromate(VI).
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i) Give the **ionic** equation for the reaction that occurs when sodium sulfate solution is added to the conical flask in Step **2**. Include state symbols.

(1)

ii) Give a possible reason why it is necessary to add sodium sulfate solution. Justify your answer.

(1)

(2 marks)

(b) Suggest why the red precipitate of silver chromate(VI) only forms after all the chloride ions have reacted.

(1 mark)

(c) Some data obtained in the experiment are shown.

Titration number	1	2	3	4
Burette reading (final) / cm³	16.15	32.05	48.30	47.40
Burette reading (initial) / cm³	0.00	16.15	32.50	31.55
Titre / cm³	16.15			

i) Complete the table and use the concordant results to calculate the mean titre.

(2)

ii) Determine the value of **x** in the formula of the hydrated salt, BaCl₂ ·**x**H₂O. Use information from the procedure and your mean titre from (c)(i).

You **must** show your working.

(5)

(7 marks)